

WHEN DOES REFERENCE MATCHING TRANSFER? SPECIFICATION-CERTIFIED AUDITS OF FIR/IIR REALIZATIONS

Xianghui Meng¹ and Jionghao Lin^{1,2,*}

¹The University of Hong Kong

²Carnegie Mellon University

margretmeng1020@gmail.com

*Corresponding author: jionghao@hku.hk

ABSTRACT

Reference-based correctness scoring treats proximity to selected realizations as a proxy for satisfaction of a digital-filter specification, even when a magnitude mask admits many legitimate FIR or IIR designs. We audit that proxy on 20 magnitude-mask tasks (16 FIR, 4 IIR) whose 412 constructed valid implementations are continuously certified, using coefficient and magnitude-response catalogs of observed valid realizations. No observed valid coefficient reference recovers exact single-center membership on any task; exact coefficient catalogs require more than 10 references on all 20 tasks (median 23). After those catalogs were frozen, a catalog-blind protocol produced 614 new continuously certified valid realizations. The frozen coefficient catalogs accepted 66/614 (10.7%; task-macro median 4.8%), whereas the frozen magnitude-response catalogs accepted 585/614 (95.3%; task-macro median 100%). Reference adequacy is therefore representation-dependent: coefficient catalogs that are exact on one finite universe under-cover later valid designs, while magnitude-response matching is a substantially better—but not universally sound—surrogate. Specification checking remains the correctness definition.

Index Terms— Digital filters, FIR, IIR, specification testing, reference matching

1. INTRODUCTION

A digital-filter design task is usually stated as a *behavior*: passband and stopband edges, ripple, attenuation, and, for IIR filters, stability [1, 2, 3]. Those constraints define a feasible set, not a unique coefficient vector. Window, frequency-sampling, least-squares, and equiripple FIR designers [4, 5, 6, 7, 8, 9], and Butterworth, Chebyshev, and elliptic IIR sections [10, 11, 12, 13], can all satisfy the same magnitude mask.

Implementation evaluation nevertheless often compares a candidate to one golden coefficient vector, one reference magnitude response, or a small catalog of library designs. That

comparison answers a different question: whether the candidate is the *same realization*, not whether it meets the specification that justified the reference. The two coincide when the specification is effectively a unique map. They need not coincide for non-unique magnitude masks.

The general test-oracle problem, specification-based conformance, and prototype or reference-set selection are established ideas [14, 15, 16, 17, 18]. We do not propose a new set-cover algorithm or a new theory of oracles. The unresolved signal-processing question is narrower: whether realization-based coefficient or magnitude-response references fitted on one FIR/IIR implementation universe remain adequate for later implementations certified against the same mask.

This paper makes three contributions. (i) We formulate correctness as membership in a specification-defined valid set $\mathcal{V}_t$ and assemble a continuously certified FIR/IIR evaluation universe. (ii) We quantify how finite catalogs of *actual* valid realizations recover that membership on the frozen universe, including an exact observed-valid catalog-complexity diagnostic. (iii) On a catalog-blind prospective challenge of 614 independently certified realizations, we show a strong representation-dependent contrast: frozen coefficient catalogs transfer poorly, frozen magnitude-response catalogs usually transfer well, and admitting the new valids substantially expands every exact coefficient catalog.

2. SPECIFICATION SETS AND REFERENCE CATALOGS

A task t asks for an FIR impulse response or an IIR pair (b, a) . A predicate $S_t(h) \in \{0, 1\}$ encodes the registered magnitude mask and, for IIR designs, a strict pole-radius bound. Correctness is membership

$$\mathcal{V}_t = \{h : S_t(h) = 1\}. \quad (1)$$

A reference catalog $\mathcal{R}$ is a finite set of *observed* specification-valid realizations. With a distance d and a common scalar

threshold τ , the acceptance set is

$$\mathcal{A}_{\tau, \mathcal{R}} = \{h : d_{\mathcal{R}}(h) \leq \tau\}, \quad d_{\mathcal{R}}(h) = \min_{r \in \mathcal{R}} d(h, r). \quad (2)$$

The reference oracle declares “correct” iff $h \in \mathcal{A}_{\tau, \mathcal{R}}$; the specification oracle declares “correct” iff $h \in \mathcal{V}_t$.

All exact-recovery statements below are restricted to an explicit finite universe $\mathcal{U}_t$. Write $V = \mathcal{V}_t \cap \mathcal{U}_t$ and $I = \mathcal{U}_t \setminus \mathcal{V}_t$, and define

$$D_V(\mathcal{R}) = \max_{v \in V} \min_{r \in \mathcal{R}} d(v, r), \quad D_I(\mathcal{R}) = \min_{i \in I} \min_{r \in \mathcal{R}} d(i, r). \quad (3)$$

Observation 1. On a frozen finite universe, a common scalar threshold recovers membership exactly if and only if $D_V(\mathcal{R}) < D_I(\mathcal{R})$. The statement is evaluation geometry, not a theorem over all filters.

The observed-valid *reference catalog complexity* K_{obs}^* is the smallest $|\mathcal{R}|$ of specification-valid occupants of V for which Observation 1 holds. Computing that finite-universe diagnostic reduces to a standard set-cover optimization [18, 17]; the optimization itself is not our contribution. K_{obs}^* measures realizable reference burden, not algorithmic novelty.

Two distances are used. Coefficient distance is a relative ℓ_2 after canonicalization (FIR: trailing-zero and sign rules; IIR: $a_0 = 1$, concatenated (b, a)). Magnitude-response distance is the RMSE of $|H|$ on the constrained pass and stop bands at $N_f = 131072$ frequency samples. Centers must be actual observed valid realizations, not arbitrary Euclidean points.

3. CERTIFIED UNIVERSE AND PROSPECTIVE PROTOCOL

The base suite comprises 20 non-unique magnitude-mask tasks: FIR low-pass, high-pass, band-pass, and band-stop at $f_s = 8$ and 16 kHz, each loose and tight (16 FIR); and IIR low-pass and high-pass at 8 kHz (4 IIR). The frozen constructed valid corpus contains 412 implementations (336 FIR, 76 IIR).

Final labels are not self-certified by the construction checker. For FIR, $P(x) = |H|^2$ with $x = \cos \omega$ is certified continuously over each constrained band by exact-rational Bernstein or Sturm sign reasoning. For IIR, we certify strict Schur stability (pole radius < 0.999) and the sign of $P_B(x) - CP_A(x)$ on every constrained band. All 412/412 constructed valids received continuous specification certification; there were no valid $\rightarrow$ invalid contradictions, and all frozen mechanism and boundary invalids certified invalid. Cosine endpoint handling is a numerical/rational certificate, not a machine-checked continuous proof. Implementation details live in the repository.

Three questions organize the audit. RQ1: when does realization-reference matching recover specification membership on this frozen universe? RQ2: how large a realizable

catalog is required for exact recovery? RQ3: do exact base-fitted catalogs transfer to later implementations that satisfy the same masks?

The prospective challenge was protocol-locked after catalog fitting. (1) Fit and freeze base coefficient and response catalogs and their max-safe thresholds (the largest τ that accepts no frozen base invalid). (2) Commit catalog identities. (3) Freeze a generation protocol that does not read those catalogs, distances, transfer scores, or ambient centers. (4) Schedule 960 valid-design attempts across all 20 tasks and eight ordinary designer families: Remez, FIRLS, frequency sampling, and windowed FIR; Butterworth, Chebyshev I, Chebyshev II, and elliptic IIR. (5) Continuously certify every occupant. (6) Remove exact coefficient duplicates. (7) Freeze 614 new nonduplicate certified-valid challenge implementations (500 FIR, 114 IIR; all 20 tasks; at least 13 per task). (8) Only then unblind reference transfer. An earlier same-order probe set participated in catalog fitting and is *not* this holdout. The 614 occupants are prospectively generated, catalog-blind, specification-certified realizations; they are not all possible filters.

4. RESULTS

4.1. Base-universe adequacy

On the frozen 412-valid universe, no single observed valid coefficient reference recovers exact membership (20/20 canonical and 20/20 best-observed non-separable). A single magnitude-response reference fails on 19/20 canonical and 18/20 best-observed tasks (Table 1). Exact observed-valid coefficient catalogs exist on 20/20 tasks, but never with $K^* \leq 10$: the median is 23 (range 15–113; median $K^*/|V| = 0.512$). Library-only catalogs recover membership on 0/20 tasks in both representations. Response K^* has median 17.5 and equals 1 on two tasks.

Arbitrary Euclidean ambient centers, which need not be realizable filters, separate 19/20 coefficient and 20/20 response finite universes (only `iir_lp_loose_8k` has no coefficient center). Poor later transfer is therefore not generally due to intrinsic one-ball inseparability of the labeled point sets; the restriction to realizable reference filters matters.

4.2. Prospective transfer

This is the primary empirical result (Fig. 1, Table 2). Frozen coefficient K_{obs}^* catalogs accepted 66/614 prospective valids (pooled 10.7%; task-macro mean 10.6%, median 4.8%; range 0–50%). Every task transferred below 75% (20/20); none reached 95%. FIR and IIR task-macro means were 11.8% and 5.5%. The same verdict is obtained under the predeclared midpoint threshold, so the coefficient failure is not an artifact of max-safe τ .

Frozen magnitude-response catalogs accepted 585/614 (pooled 95.3%; task-macro mean 94.6%, median 100%).

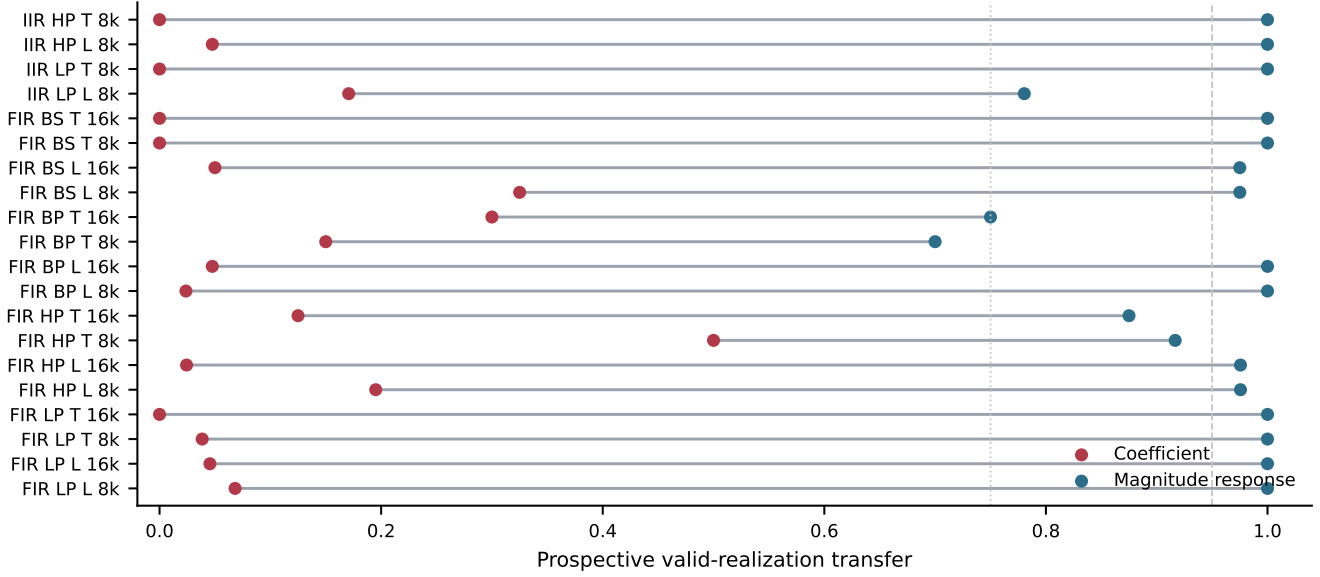


Fig. 1. Catalog-blind prospective transfer on 614 specification-certified realizations. Each row is one magnitude-mask task (L/T: loose/tight). Red: frozen coefficient catalogs. Blue: frozen magnitude-response catalogs. Guides mark 75% and 95%.

Table 1. Exact recovery on the frozen base universe. “Non-sep.” is a finite-universe single-center failure. K^* is the minimum observed-valid catalog size.

Repr.	Can. non-sep.	Best obs.	med. K^*	Lib. exact
Coeff.	20/20	20/20	23	0/20
Resp.	19/20	18/20	17.5	0/20

Fifteen tasks reached at least 95%; one tight FIR band-pass task transferred at 70%. FIR and IIR macros were both about 94.6%. Response transfer is generally strong and still task-specific.

The coefficient result is not driven by one construction family: Remez, FIRLS, frequency-sampling, and windowed FIR transferred at 13.8%, 13.0%, 11.5%, and 5.1%; Butterworth, Chebyshev I, II, and elliptic IIR transferred at 18.2%, 3.6%, 6.9%, and 5.7%. Response transfer for those families ranged from 85.7% to 100%.

Enlarging the coefficient catalog on the *base* universe does not rescue prospective transfer. Canonical $K=1$, published $K \in \{3, 5\}$, and all-library coefficient oracles accepted 0/614; the best observed single coefficient reference accepted 7.5%; the exact K_{obs}^* catalog accepted 10.7%. The corresponding response figures are 38.9% (canonical) and 95.3% (K_{obs}^*). Increasing coefficient-catalog size enough to fit the original finite universe therefore does not produce strong prospective coefficient transfer.

Table 2. Prospective transfer of frozen base catalogs to 614 catalog-blind certified-valid realizations.

Repr.	Accepted	Pooled	Macro med.	$\geq 95\%$	$< 75\%$
Coeff.	66/614	10.7%	4.8%	0/20	20/20
Resp.	585/614	95.3%	100%	15/20	1/20

4.3. Catalog maintenance

Admitting the 614 prospective valids and recomputing exact coefficient catalogs raises the median K^* from 23 to 55 (median relative growth 52.5%). Every task has $M^* > 0$: every minimum-size expanded coefficient catalog requires at least one newly admitted realization (median $M^* = 15.5$, maximum 37). In-sample catalog exactness therefore does not eliminate later support-expansion burden. Response catalogs grow much less (median $\Delta K = 0$; $M^* > 0$ on 9/20 tasks).

5. DISCUSSION

That Remez, window, and elliptic designs can meet one mask with different coefficients is classical filter theory; the audit is whether a catalog of those realizations remains an adequate correctness proxy for later valid designs. Coefficient proximity records implementation choice: distinct constructions that meet the same mask can sit far apart in (b, a) space. A catalog that is exact on one finite sample of those constructions can still miss later legitimate occupants. Magnitude-response proximity discards much of that realization-specific variation and therefore usually transfers. It remains an ap-

proximation to S_t : it failed on one task in Fig. 1, and it is not a proof of mask satisfaction.

A golden coefficient vector, or a finite coefficient catalog fitted to one implementation corpus, should not be read as a correctness oracle for a non-unique filter-design task without an adequacy audit. Where magnitude-response matching transfers, it can be a useful surrogate. The criterion used to define correctness in this study is still S_t .

The study is finite. We evaluate 20 magnitude-mask tasks, not general DSP software. The prospective challenge uses eight standard designer families, not all implementations. A secondary invalid set of 310 occupants covers only 12/20 tasks; on that incomplete set, coefficient catalogs falsely accepted 38/310 implementations and magnitude-response catalogs 96/310. Those rates are not a 20-task false-acceptance rate. Ambient Euclidean centers often exist, so the geometry is not intrinsic one-center inseparability. No infinite-universe

claim is made, and magnitude-response matching is not asserted to be sound or complete in general.

6. CONCLUSION

Reference matching should be audited as a surrogate for a digital-filter specification, not assumed equivalent to it. Across 614 prospectively generated valid FIR/IIR realizations, coefficient catalogs that were exact on one implementation universe transferred poorly, while magnitude-response catalogs were substantially more stable. Correctness evaluation should distinguish specification compliance from realization identity and choose the reference representation accordingly.

Frozen specifications, certified corpora, protocol locks, and a reproduction command are provided in the accompanying repository.

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7. REFERENCES

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